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Portfolio · Selected Projects · 2019 – 2026

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Overview

A selection of products I have built or contributed to — from **production SaaS used by hundreds of companies** to **apps published on Google Play**, **AI agents and on-device AI**, and **systems and numerical-computing** side projects. Most of it is **TypeScript end to end** (Next.js, tRPC, Prisma, React Native/Expo), with excursions into **Kotlin, Rust/WebAssembly, Go and Python** where the problem called for it. Each entry lists my role, the problem, what I built and the stack; public repositories are under **github.com/gitnlsn**.

200+	1M+	3	25+
companies on Toki	invoices in Zuppy	apps on Google Play	projects shipped

Production Products

Commercial software built as part of a product team, in production with paying customers.

Afya Intelligence — AI agents for medical education 2026

HubXP, for Afya ♦ Contributor

- Built a **speech-to-speech OSCE interview simulator**: a voice AI agent that plays the patient in an Objective Structured Clinical Examination so medical students can practice clinical interviewing out loud.
- Built the **virtual-patient configuration CRUD** — a multi-field form for creating and editing the patient cases the simulator plays.
- Implemented a **real-time speech-to-speech tutor agent** with working **tool calls** that answers students' questions by querying Afya's **Whitebook** clinical-reference database.

Toki — project & billing SaaS for construction companies 2023 – 2026

HubXP ♦ Full-stack developer, web + mobile ♦ 1,200+ commits

- SaaS for construction and renovation companies to organize project data, media and billing; onboarded **200+ companies** and **1,000+ React Native installs**.
- Turborepo monorepo with **Next.js** web, **Expo / React Native** mobile and a shared **tRPC + Prisma** API — type safety from database schema to UI.
- Integrated the **Iugu payment gateway** for in-app billing — later a dual integration with backward compatibility; built CNPJ company autocomplete and moved PDF generation to Vercel functions.

Stack: TypeScript, Next.js, React Native (Expo), tRPC, Prisma, PostgreSQL, Iugu, Vercel, EAS

Zuppy — electronic invoicing (*nota fiscal*) for digital-product sellers 2023 – 2025

HubXP ♦ *Full-stack developer* ♦ *560+ commits*

- Fiscal-domain platform that issues Brazilian electronic invoices at scale for *infoprodutores*, integrated with the **plugnotas** provider to emit invoices reliably at scale.
- Rewrote the invoice search query, cutting response time from **~2 min to ~3 s (~40×)** on a table past **1M records**; added **Sentry** and **PostHog** for observability.
- Prototyped **Zuppy AI**, a set of **Mastra** agents that onboard a company conversationally and generate subscription checkout links, with Zod-validated tool outputs.

Stack: TypeScript, Next.js, tRPC, Prisma, PostgreSQL, plugnotas, Mastra, Vitest, Sentry, PostHog

Talent Discovery — AI agent that sources engineers from GitHub 2026

HubXP ♦ *Author*

- Given a job description in Markdown, searches GitHub via **GraphQL**, enriches profiles (languages, contributions, repositories) and ranks candidates on a **0–100 multi-dimensional score** with confidence levels.
- "Code over clout": scoring weighs PR descriptions, review quality and documentation over follower counts; results cached and exported as JSON, CSV or table.

Stack: TypeScript, Mastra, Gemini, GitHub GraphQL API, Node.js

HubXP Institutional & Talent Marketplace 2022 – 2026

HubXP ♦ *Contributor*

- Company website and talent marketplace on the T3 stack; Next.js pages → tRPC procedures → **use-case layer** → Prisma, with Better Auth and Dockerized local environment.

Stack: TypeScript, Next.js, tRPC, Prisma, PostgreSQL, Better Auth, Docker

Published Mobile Apps

Apps I designed, built and published myself on Google Play.

Voile Drift — AI-powered calendar insights Google Play

Solo ♦ *play.google.com/store/apps/details?id=com.voiledrift.mobile*

- Logs the user's events with rich descriptions and runs **Gemini Flash** over them to surface relationships between events and generate insights.
- Shipped with its own Next.js landing page and a CLI that generates Instagram carousel cards for marketing.

Stack: Kotlin, Android, Gemini Flash API, Next.js (landing page)

Vera — menstrual cycle forecasting Google Play

Solo ♦ *play.google.com/store/apps/details?id=com.nelson.vera.mobile*

- Predicts and tracks menstrual cycles from user-logged history, fully on-device.

Stack: Kotlin, Android

Lontra Ágil — AI study companion for *concursos públicos* Google Play 2025 – 2026

Team project ♦ *Mobile app lead* ♦ *play.google.com/store/apps/details?id=com.nllsn.pubconapp*

- Imports public-exam PDFs, uses **Gemini Flash** to parse them and auto-generates practice questions for students preparing for public-sector entrance exams.
- Built the **Expo development-build** mobile app and the first T3 web version; the platform was later rebuilt as a Turborepo monorepo.

Stack: TypeScript, React Native (Expo), Next.js, tRPC, Prisma, Gemini Flash API

NLSN Meditate — guided meditation with progress and leaderboards 2026

Solo ♦ *Android*

- Guided meditations, a lesson library on meditation and its science, streaks and badges, wired to **Google Play Games** achievements and leaderboards.

Stack: TypeScript, React Native (Expo), EAS, Google Play Games Services

AI & Agents

LLM-powered tools — from cloud agents to models running entirely in the browser.

CNPJ Discovery — B2B lead qualification from Brazilian open data 2026 Open source

Solo ♦ *github.com/gitnlsn/cnpj-discovery*

- Turns an ideal-customer profile into qualified leads: an **LLM suggests target CNAE codes**, the app filters every active company in the Receita Federal dataset, crawls their websites and scores them against a compiled rubric.
- Streams the monthly **2.9 GB** public dump through an in-flight filter into **Parquet + DuckDB** — rows that fail the filter never touch disk; runs at **R$ 0** using free Gemini/OpenRouter tiers.

Stack: TypeScript, Next.js, DuckDB, Parquet, Gemini / OpenRouter, pnpm workspaces

Viva Voz — read-aloud practice for French and Spanish 2026 Open source

Solo ♦ *github.com/gitnlsn/language-practice*

- Generates CEFR-levelled sentences with Chrome's built-in **Gemini Nano (Prompt API)**, translates with the **Translator API** and reads them with **Kokoro-82M** on WebGPU — no backend and no API keys.
- Compiled **eSpeak NG to WASM** for French/Spanish phonemization; history and saved sentences persist in IndexedDB; printable handwriting sheets.

Stack: TypeScript, Vite, Chrome Built-in AI, kokoro-js, ONNX / WebGPU, WASM

Ms. Ada — speech-to-speech English tutor in the browser 2026

Solo

- Spoken classes organized by topic: listens with the Web Speech API, answers and corrects with **Gemini Nano** or **WebLLM** (Llama 3.2 / Qwen 2.5 on WebGPU) behind a common provider interface.
- Zero running cost and fully private — static prerendered site, conversations never leave the machine; detects unsupported browsers before offering a 2 GB model download.

Stack: TypeScript, Next.js, WebLLM, Chrome Prompt API, Web Speech API

Lumière — English lessons by phone with an AI teacher 2025

Solo

- Scheduled 60-minute phone conversations with an emotionally-aware AI tutor, with Google Calendar scheduling, per-lesson payments, transcripts and pronunciation scoring.

Stack: TypeScript, Next.js 15, Hume AI (EVI), Twilio, Tailwind CSS

Tucano Typing — typing practice with AI-generated text 2025 Open source

Solo ♦ *github.com/gitnlsn/tucano-typing*

- Captures every keystroke with timestamps to compute WPM and accuracy, finds commonly mistyped words and generates personalized practice text; public/private scoreboards.

Stack: TypeScript, Next.js, tRPC, Prisma, NextAuth, LLM text generation

Full-Stack Platforms

Savoir Link — local-services marketplace with payments and payouts 2025 – 2026 Open source

Solo ♦ *github.com/gitnlsn/nlsn-services-marketplace*

- Customers discover a provider, book, chat in real time and pay by card or **Pix**; providers manage catalog, availability and withdrawals. Booking lifecycle modeled as a **state machine** with review gates.
- **Pagar.me** webhooks, refunds and KYC-verified provider payouts; **Ably** messaging; 26-model Prisma schema; **109 test files** across unit, integration and Playwright E2E.
- Follow-up **reverse-marketplace** variant (leads model): customers pay to post an order, providers buy credits to unlock contacts.

Stack: TypeScript (strict), Next.js 15, React 19, tRPC v11, Prisma, PostgreSQL, NextAuth v5, Pagar.me, Ably, Resend, Vitest, Playwright

Pulse FX — BRL exchange-rate and macro indicators dashboard 2026 Open source

Technical challenge ♦ *github.com/gitnlsn/pulse-fx*

- Syncs public data from the **Central Bank of Brazil** (PTAX/SGS) and **FRED** into Postgres and serves it through its own API and a React SPA.
- Layered **SOLID** backend (routes → controllers → services → repositories); data sources behind a Strategy/Adapter interface; one shared package holds DTOs and the variation rule used by both web and API.

Stack: TypeScript, Fastify, Prisma, PostgreSQL, Vite, React, Zod, Docker Compose

Engineering & Systems

Projects where the interesting part is the engine underneath.

FEA Online — finite-element analysis in the browser 2026 Open source

Solo ♦ *github.com/gitnlsn/fea-online*

- Draw a geometry, mesh it and solve on it — all client-side. The mesher and a **discontinuous Galerkin solver** are written in **Rust** and compiled to **WebAssembly**, running in a Web Worker.
- Solves steady heat diffusion with per-edge boundary conditions (plotted in 2D or as a 3D surface) and **transient compressible Euler** — shock waves reflecting and diffracting around the drawn geometry.
- The mesher, **nlsn-delaunay-refine** (constrained Delaunay triangulation and refinement), is published separately on **crates.io**.

Stack: Rust, WebAssembly (wasm-bindgen), TypeScript, Next.js, Web Workers, Canvas

Eleven Deep — football management simulation 2026 Open source

Solo ♦ *github.com/gitnlsn/game-1*

- Simulation engine as a **pure TypeScript package** (world generation, match engine, league, economy) that runs identically in a CLI tuning harness and on-device.
- Match engine **calibrated against real benchmarks** across 40-season runs, with a calibration check that fails the build on drift; **284 tests**.

Stack: TypeScript, React Native (Expo), React Navigation, Vitest

TEA App — evidence-based activities for autistic children 2026

Solo

- Offline activity library for a child and a supporting adult, where every design decision traces to a paper: built from a corpus of **67 open-access studies**, each activity carries an in-app card stating the evidence and its caveats.
- Ten activities (emotion recognition, routines, cooperative play, rhythm via accelerometer, pattern reasoning…) plus a sensory profile that configures the whole app; automated checks for contrast, offline-safety and navigation, **118 tests**.

Stack: TypeScript, React Native (Expo), Jest

Time-Off Microservice — balance integrity against an external HCM 2026 Open source

Technical challenge ♦ *github.com/gitnlsn/wizdaa-time-off*

- Request lifecycle, local balance cache with a **total / reserved / deducted** model, and a sync engine that detects drift from the HCM and flags affected requests for revalidation; `BEGIN IMMEDIATE` transactions prevent double-booking.

Stack: TypeScript, NestJS 11, SQLite, Jest, Supertest

Lead Management & AI Classification — async backend 2026 Open source

Technical challenge ♦ *github.com/gitnlsn/quark-backend_challenge*

- Leads enriched through an external API and classified by a local LLM (**Ollama**), with the work processed asynchronously through **RabbitMQ**; full stack brought up by Docker Compose.

Stack: TypeScript, NestJS 11, Prisma, PostgreSQL, RabbitMQ, Ollama, Vitest, Docker

Network & camera security monitors 2025 Open source

Solo ♦ *github.com/gitnlsn/veillle-shell*

- **PCAP Monitor**: Go packet capture detecting DNS hijacking and SSL stripping, multi-path verification over VPN/Tor/proxy routes, plus a Python deception engine and Dockerized honeypot.
- **Roof Camera Monitor**: Go service holding redundant RTSP connections, comparing audio fingerprints across streams to detect tampering, auto-failover on stalls and YouTube upload of anomalous segments; live TUI dashboard.

Stack: Go, Python, Docker, ffmpeg, YouTube Data API

Collaborations & Smaller Builds

Animalesko — pet adoption & services platform
Consumer app, provider back-office and landing page sharing one Postgres, tRPC API and design system. Team project, contributor.
Next.js, Expo, tRPC, Prisma, Better Auth, Turborepo

Benevolus — zero-fee donation platform
Open-source charity campaign platform with Pix and Stripe Connect payments. Contributor.
Laravel, Next.js, HeroUI, Woovi, Stripe, Terraform, AWS

Sevenpass — ticketing marketplace for experiences
Search, listings and secure ticket checkout for events and festivals. Contributor.
Next.js, tRPC, Prisma, NextAuth, PlanetScale, Upstash Redis

Neo-news — classifieds platform
Create, manage and search listings with Google sign-in.
Next.js, tRPC, Prisma, Drizzle, NextAuth

Gesture Controller — webcam slide navigator
Hand-gesture recognition in the browser to drive a slide deck hands-free.
Next.js, MediaPipe

Twitch Stream JS — interactive stream games
Chess and trivia rendered server-side and streamed live via native ffmpeg.
Node.js, TypeScript, ffmpeg

Google Maps Crawler — place data extraction CLI
Searches places, scrolls results and extracts ratings, reviews and addresses.
TypeScript, Puppeteer, yargs

Client websites — institutional sites & landing pages
Sites for a Tibetan Buddhist temple, a home-décor studio, an online meditation course, a mindset program and the Voile Drift launch.
Next.js, Tailwind CSS, Vercel